Day 5: Case Study Practical (1 hour): Work on a scenario involving multiple hardware issues.

1.A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

ANS:

1. Check the socket: Make sure the power socket is working.

2. Check the charger: Look for broken wires or damage.

3. Check the charging light: See if the laptop's charging light turns on.

4. Check the battery: Look for battery damage or swelling.

5. Restart the laptop: Unplug the charger, hold the power button for 15 seconds, reconnect the charger, and try turning it on.

**2.You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.**

ANS: Step 1: Check the monitor.
Make sure the monitor is switched on and the power light is on.

Step 2: Check the power cable.
Make sure the monitor's power cable is connected properly.

Step 3: Check the display cable.
Check that the HDMI or VGA cable is connected properly to the monitor and PC.

Step 4: Check the input source.
Select the correct input, such as HDMI or VGA, on the monitor.

Step 5: Restart the PC.
Turn off the PC and monitor, wait a few seconds, and turn them on again.

Step 6: Check the brightness.
Make sure the monitor brightness is not set too low.

Step 7: Check the RAM.
If the screen is still blank, turn off and unplug the PC, then check whether the RAM is fitted properly.

Step 8: Check the graphics card.
Make sure the graphics card and its power cables are connected properly.

Step 9: Test another monitor.
Connect a working monitor to check whether the original monitor is faulty**.**

3.A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

ANS:

Step 1: Check the battery.
Replace the mouse battery or charge it.

Step 2: Check the mouse switch.
Make sure the mouse is turned on.

Step 3: Check the USB receiver.
Remove the receiver and plug it into another USB port.

Step 4: Test another mouse.
Connect another working mouse to check whether the problem is with the original mouse.

Step 5: Check system settings.
Open Windows Settings and check whether the mouse is detected and working properly.

Step 6: Restart the computer.
Restart the PC and check whether the mouse works.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.

ANS:

1. Faulty Fan

- Cause: The CPU or cabinet fan is not working.
- Check: See if the fan is spinning when the PC is running.
- Solution: Replace the faulty fan.

2. Dust Inside the PC

- Cause: Dust blocks airflow and makes the PC hot.

- Check: Open the PC case and look for dust on fans and vents.
- Solution: Clean the dust using compressed air while the PC is unplugged.

3. Dry Thermal Paste

- Cause: Old thermal paste cannot transfer heat properly from the CPU to the cooler.
- Check: Inspect the thermal paste between the CPU and cooler.
- Solution: Clean off the old paste and apply new thermal paste correctly.